Green Bonds EDA

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```r
# Load in all libraries
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.3

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.1     v stringr   1.5.1
## v lubridate 1.9.4     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(readxl)
library(ggplot2)
library(maps)
```

```
## Warning: package 'maps' was built under R version 4.4.3

##
## Attaching package: 'maps'
##
## The following object is masked from 'package:purrr':
##
##     map
```

```r
library(tidyr)
```

```r
# Solar Data
solar_2024 <- read_excel("Solar.xlsx", sheet = "2024")
solar_2023 <- read_excel("Solar.xlsx", sheet = "2023")
solar_2022 <- read_excel("Solar.xlsx", sheet = "2022")
solar_2021 <- read_excel("Solar.xlsx", sheet = "2021")
solar_2020 <- read_excel("Solar.xlsx", sheet = "2020")
solar_2019 <- read_excel("Solar.xlsx", sheet = "2019")
solar_2018 <- read_excel("Solar.xlsx", sheet = "2018")
solar_2017 <- read_excel("Solar.xlsx", sheet = "2017")
solar_2016 <- read_excel("Solar.xlsx", sheet = "2016")
solar_2015 <- read_excel("Solar.xlsx", sheet = "2015")
solar_2014 <- read_excel("Solar.xlsx", sheet = "2014")
solar_2013 <- read_excel("Solar.xlsx", sheet = "2013")

# Wind Data
wind_2024 <- read_excel("Wind.xlsx", sheet = "2024")
wind_2023 <- read_excel("Wind.xlsx", sheet = "2023")
wind_2022 <- read_excel("Wind.xlsx", sheet = "2022")
wind_2021 <- read_excel("Wind.xlsx", sheet = "2021")
wind_2020 <- read_excel("Wind.xlsx", sheet = "2020")
wind_2019 <- read_excel("Wind.xlsx", sheet = "2019")
wind_2018 <- read_excel("Wind.xlsx", sheet = "2018")
wind_2017 <- read_excel("Wind.xlsx", sheet = "2017")
wind_2016 <- read_excel("Wind.xlsx", sheet = "2016")
wind_2015 <- read_excel("Wind.xlsx", sheet = "2015")
wind_2014 <- read_excel("Wind.xlsx", sheet = "2014")
wind_2013 <- read_excel("Wind.xlsx", sheet = "2013")

# Green Bond Issuance and RPS Data
GBI <- read_excel("GB_RPS Hard Code.xlsx")

# Pre-Trends Wind Data
pre_wind <- read_csv("Wind Pre.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 151457 Columns: 28
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (11): case_id, faa_ors, faa_asn, t_state, t_county, t_fips, p_name, t_ma...
## dbl (17): usgs_pr_id, eia_id, p_year, p_tnum, p_cap, t_cap, t_hh, t_rd, t_rs...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Pre-Trends Solar Data
pre_solar <- read_excel("Solar 2010 - 2012.xlsx")
```

```r
# Cleaning Pre-Wind Data

clean_wind_panel <- pre_wind %>%
# Remove any rows where the operational year is missing/blank
filter(!is.na(p_year)) %>%

# Select only the specific columns your model needs
select(t_state, p_year, t_cap) %>%

# Group by State and Year, and convert individual turbine kW to total MW added
group_by(t_state, p_year) %>%
summarize(mw_added = sum(t_cap, na.rm = TRUE) / 1000, .groups = 'drop') %>%

# Open your 50-state grid for your pre-trends window
complete(t_state, p_year = 2000:2026, fill = list(mw_added = 0)) %>%
arrange(t_state, p_year) %>%
group_by(t_state) %>%

# Cumulative sum holds capacity at 0 until first project,
# then holds total capacity flat forward through the years
mutate(total_wind_mw = cumsum(mw_added)) %>%
rename(state = t_state, year = p_year) %>%

# Filter the outcome variable to strictly include 2010-2012
filter(year >= 2010 & year <= 2012)
```

```r
# Cleaning Pre-Solar Data

# 2. Extract columns by their exact position number instead of text names
solar_wide_cumulative <- pre_solar %>%
  # Select column 2 (State), column 3 (2010), column 4 (2011), and column 5 (2012)
  select(State = 1, 2, 3, 4)

# 3. Check what R actually named these columns so we can use them
col_names <- colnames(solar_wide_cumulative)

# 4. Perform the math using index tracking to prevent text crashes
solar_wide_cumulative <- solar_wide_cumulative %>%
  mutate(
    Year_2010 = .[[2]],                  # Pulls the data from the 2010 column position
    Year_2011 = .[[2]] + .[[3]],         # Adds 2010 + 2011 column positions
    Year_2012 = (.[[2]] + .[[3]]) + .[[4]] # Adds 2010 + 2011 + 2012 column positions
  ) %>%

  # 5. Output only your State and the clean, quote-free cumulative totals
  select(State, Year_2010, Year_2011, Year_2012)
```

```r
# Vector of years
years <- 2024:2013
print("hello")
```

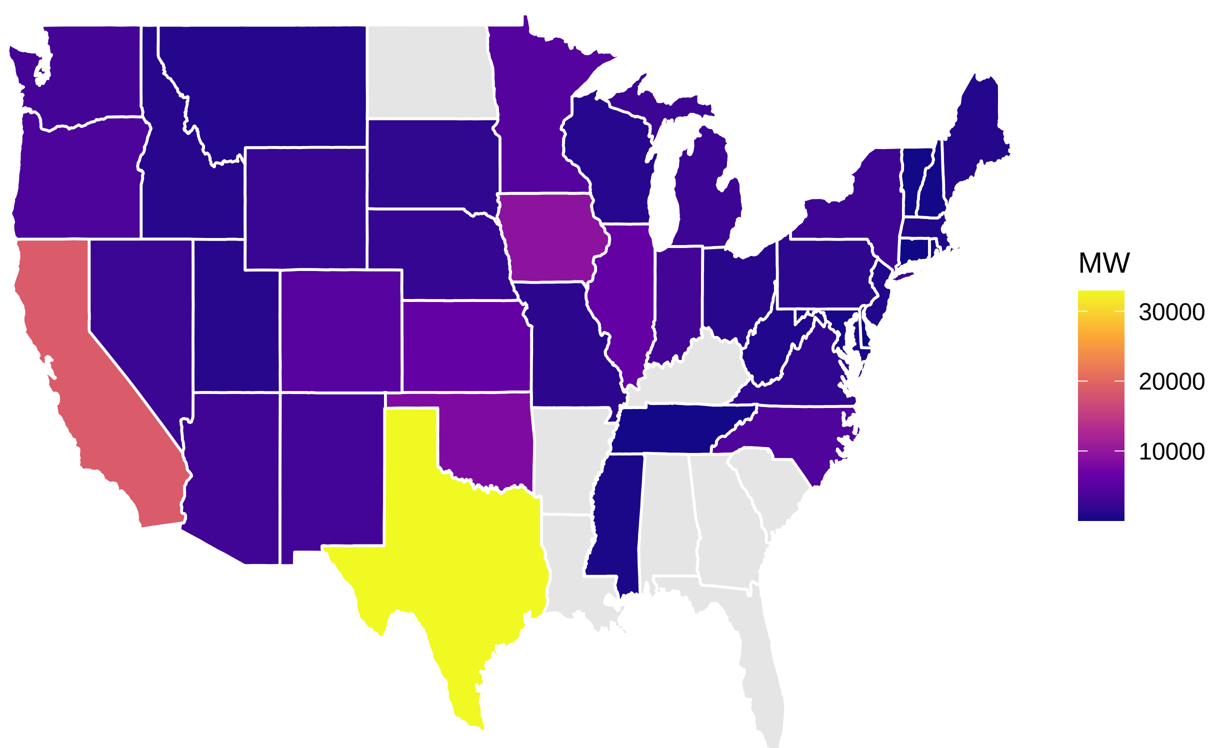
```
## [1] "hello"
```

```r
# Initialize empty list to store per-year state totals
solar_totals <- list()

for (yr in years) {
  # Get the dataframe for that year using get()
  df <- get(paste0("solar_", yr))

  # Sum capacity per state
  solar_totals[[as.character(yr)]] <- df %>%
    group_by(State) %>%
    summarise(TotalCapacity = sum(`Nameplate Capacity (MW)`, na.rm = TRUE))
}

solar_all_years <- bind_rows(
  lapply(names(solar_totals), function(yr) {
    solar_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
)

# Initialize empty list to store per-year state totals for wind
wind_totals <- list()

for (yr in years) {
  # Get the dataframe for that year dynamically
  df <- get(paste0("wind_", yr))

  # Sum capacity per state
  wind_totals[[as.character(yr)]] <- df %>%
    group_by(State) %>%
    summarise(TotalCapacity = sum(`Nameplate Capacity (MW)`, na.rm = TRUE))
}

wind_all_years <- bind_rows(
  lapply(names(wind_totals), function(yr) {
    wind_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
)
```

```r
solar_mean <- bind_rows(
  lapply(names(solar_totals), function(yr) {
    solar_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
) %>%
  group_by(State) %>%
  summarise(MeanSolar = mean(TotalCapacity, na.rm = TRUE))

wind_mean <- bind_rows(
  lapply(names(wind_totals), function(yr) {
    wind_totals[[yr]] %>% mutate(Year = as.numeric(yr))
  })
) %>%
  group_by(State) %>%
  summarise(MeanWind = mean(TotalCapacity, na.rm = TRUE))
```

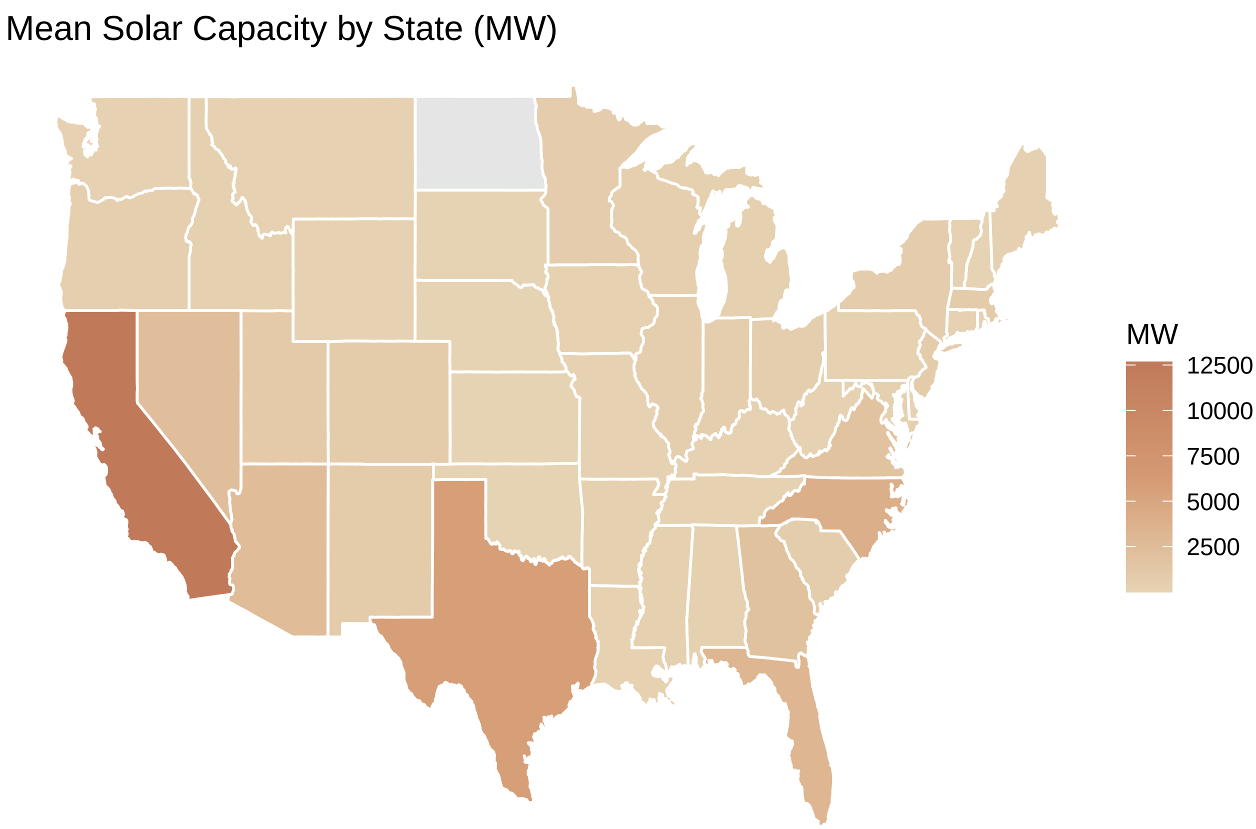
```r
# Merge solar and wind means
renewables_mean <- full_join(solar_mean, wind_mean, by = "State") %>%
  mutate(MeanRenewable = MeanSolar + MeanWind)

us_states <- map_data("state")

# Make sure state names match: lowercase
renewables_mean$region <- tolower(state.abb[match(renewables_mean$State, state.abb)])

# Convert state abbreviations to lowercase full names
renewables_mean$region <- tolower(state.name[match(renewables_mean$State, state.abb)])

us_map_data <- left_join(us_states, renewables_mean, by = "region")

ggplot(us_map_data, aes(x = long, y = lat, group = group, fill = MeanRenewable)) +
  geom_polygon(color = "white") +
  scale_fill_viridis_c(option = "C", na.value = "grey90") +
  labs(
    title = "Mean Renewable Energy Capacity by State (MW)",
    fill = "MW"
  ) +
  theme_void() +
  theme(legend.position = "right")
```

Mean Renewable Energy Capacity by State (MW)



```r
# Map data
us_states <- map_data("state")

# Merge map with solar and wind means separately
solar_map_data <- left_join(us_states, renewables_mean %>% select(region, MeanSolar), by = "region")
wind_map_data  <- left_join(us_states, renewables_mean %>% select(region, MeanWind), by = "region")
```

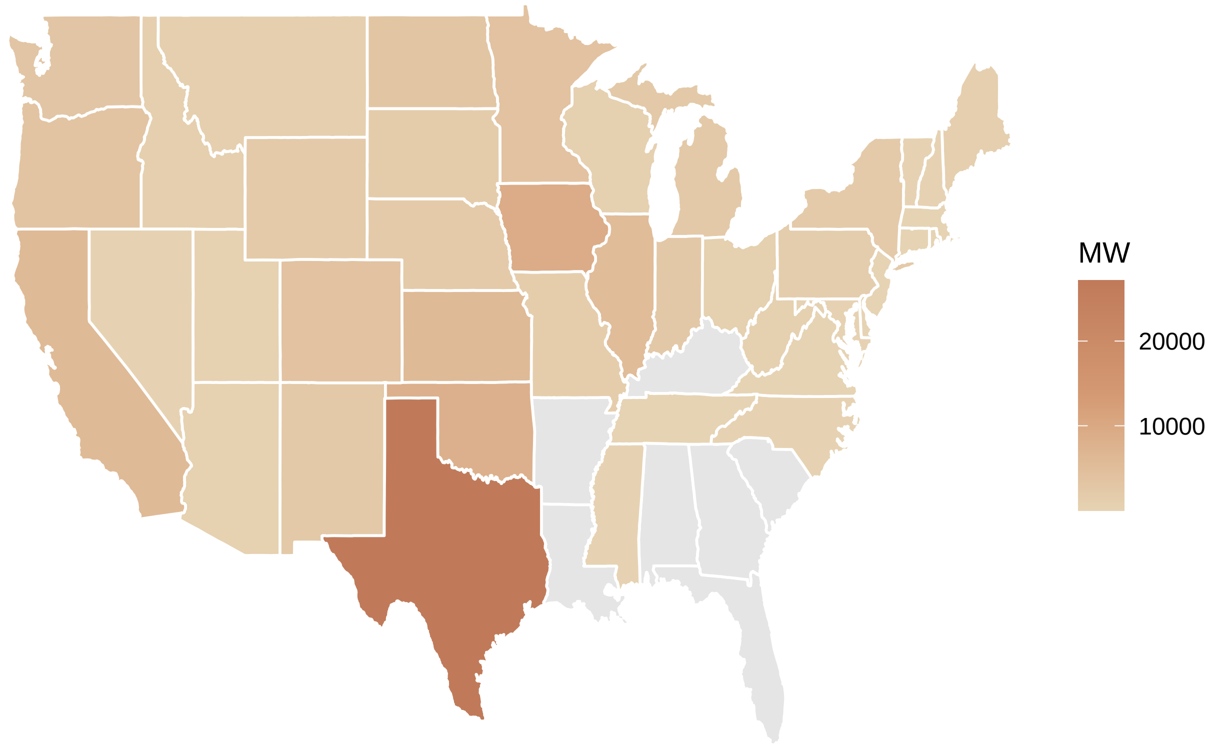
```r
solar = ggplot(solar_map_data, aes(x = long, y = lat, group = group, fill = MeanSolar)) +
  geom_polygon(color = "white") +
  scale_fill_gradientn(
    colors = c(
      "#E6D3B3",
      "#D49A73",
      "#C07A5A"
    ),
    na.value = "grey90"
  ) +
  labs(
    title = "Mean Solar Capacity by State (MW)",
    fill = "MW"
  ) +
  theme_void() +
  theme(
    legend.position = "right",
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA)
  )

print(solar)
```




```
ggsave("solar.png", plot = solar, width = 6, height = 4, dpi = 300)
```

```
wind = ggplot(wind_map_data, aes(x = long, y = lat, group = group, fill = MeanWind)) +
  geom_polygon(color = "white") +
  scale_fill_gradientn(
    colors = c(
      "#E6D3B3",
      "#D49A73",
      "#C07A5A"
    ),
    na.value = "grey90"
  ) +
  labs(
    title = "Mean Wind Capacity by State (MW)",
    fill = "MW"
  ) +
  theme_void() +
  theme(
    legend.position = "right",
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA)
  )

print(wind)
```

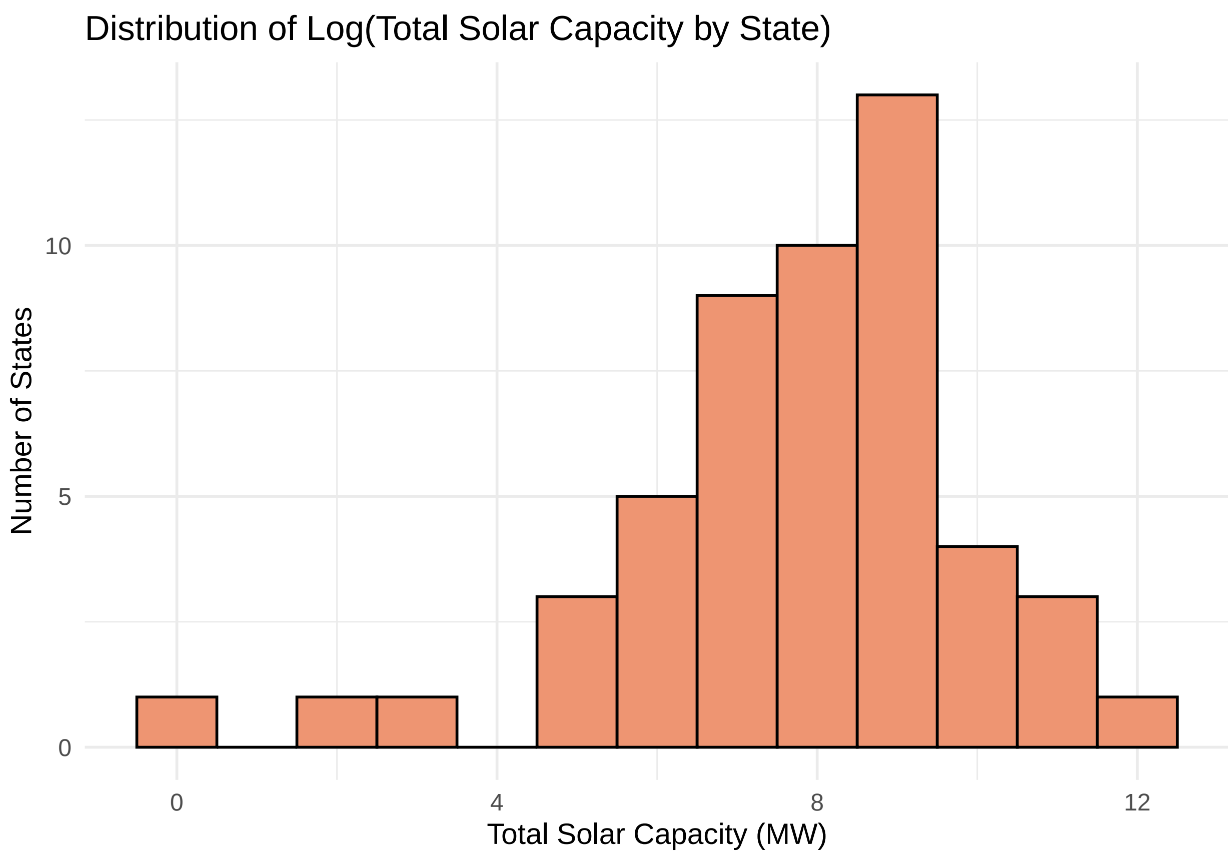
Mean Wind Capacity by State (MW)



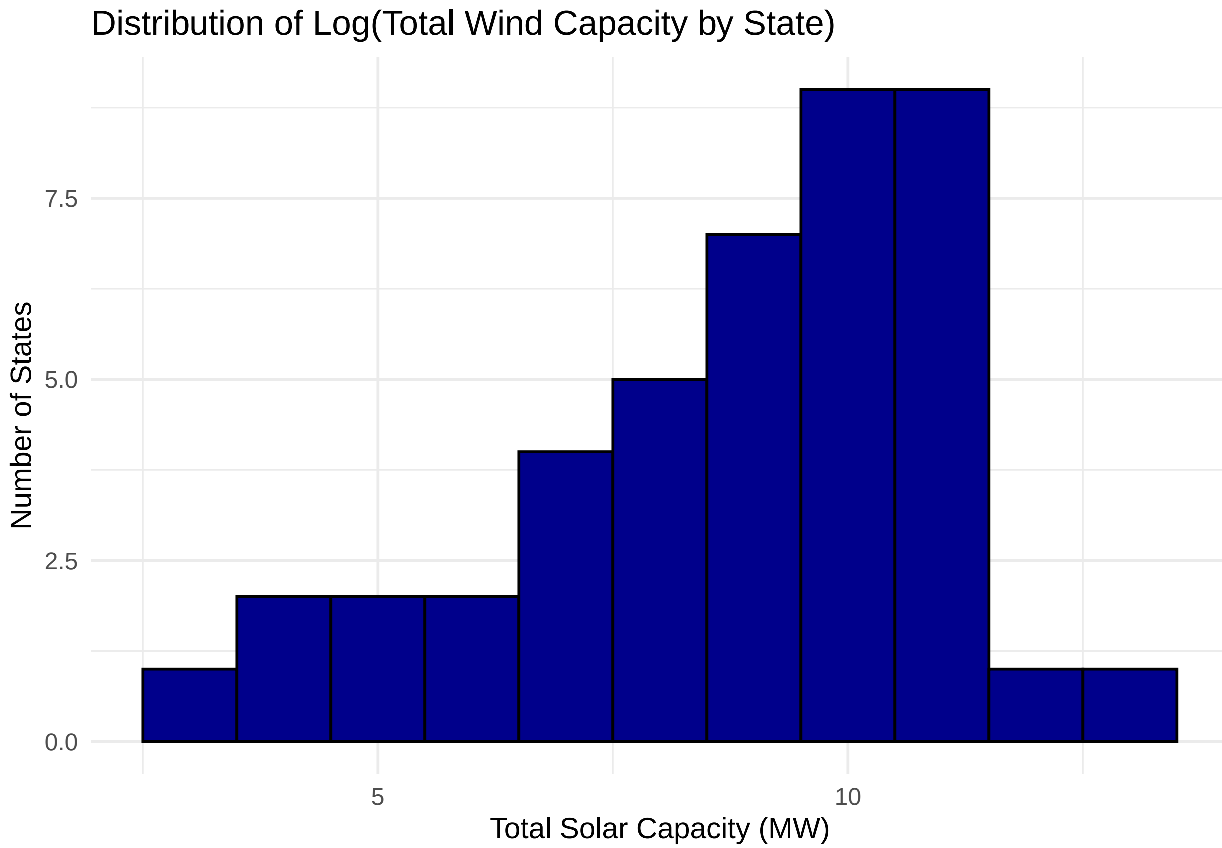

```r
ggsave("wind.png", plot = wind, width = 6, height = 4, dpi = 300)
```

```r
# Sum total capacity per state across all years
solar_total_state <- solar_all_years %>%
  group_by(State) %>%
  summarise(TotalCapacity = sum(TotalCapacity, na.rm = TRUE))

wind_total_state <- wind_all_years %>%
  group_by(State) %>%
  summarise(TotalCapacity = sum(TotalCapacity, na.rm = TRUE))
```

```r
ggplot(solar_total_state, aes(x = log(TotalCapacity))) +
  geom_histogram(binwidth = 1, fill = "lightsalmon2", color = "black") +
  labs(
    title = "Distribution of Log(Total Solar Capacity by State)",
    x = "Total Solar Capacity (MW)",
    y = "Number of States"
  ) +
  theme_minimal()
```




```r
ggplot(wind_total_state, aes(x = log(TotalCapacity))) +
  geom_histogram(binwidth = 1, fill = "darkblue", color = "black") +
  labs(
    title = "Distribution of Log(Total Wind Capacity by State)",
    x = "Total Solar Capacity (MW)",
    y = "Number of States"
  ) +
  theme_minimal()
```


Distribution of Log(Total Wind Capacity by State)
Number of States
7.5
5.0
2.5
0.0
5
10
Total Solar Capacity (MW)